

Focus Ineffability Effects Follow from Conditions on Question Answering

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Abstract

We examine an ineffability puzzle first identified by Schwarzschild [1993], in which a question containing a focus-sensitive operator cannot be felicitously answered, regardless of focus assignment. Previous accounts attribute the problem to a ban on crossing dependencies between focus operators and their associates; we instead derive it from the interaction between semantic content and general pragmatic considerations. We analyze ineffability within a question-based approach, according to which focus in an assertion corresponds to the implicit question it addresses. In ineffability examples, the same focus-sensitive operator occurs in every member of the implicit question denotation. As a result, the operator’s domain of quantification influences the question signaled by focus; and in the relevant cases, no admissible choice of domain yields an interpretation that matches the explicit, contextually-supplied question. We further relate ineffability to wh-intervention effects [Beck, 1996], arguing that intervention constructions likewise involve a single operator whose domain is shared across question alternatives. Finally, we show that the present account improves on earlier approaches that rule out crossing dependencies categorically: beyond being less stipulative, it correctly predicts that ineffability can be obviated by manipulating the preceding question and other contextual factors.

1 Focus ineffability

1.1 The puzzle

Schwarzschild [1993, 1997, 2004, 2020] observes that a question like (1) cannot be answered using either (1-a) or (1-b):

- (1) What does John only eat in PARIS?
a. # John only eats CREPES in Paris.
b. # John only eats crepes in PARIS.

The question is perfectly understandable, yet there seems to be no way of assigning focus that would yield an acceptable answer:¹ (1-a) gives rise to the unwanted inference that John eats nothing but

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crepes when in Paris (rather than conveying that he eats crepes in Paris and nowhere else), whereas (1-b) is understood as answering the wrong question, namely *Where does John eat crepes?*

Example (1) exhibits focus ineffability, that is, the inability to assign focus to a particular given structure (there are, of course, alternative structural means of conveying the relevant answer meaning, to which we return in Section 3.1). Cases of apparent grammatical breakdown are often revealing about the underlying mechanisms; therefore, a closer examination of this example may provide some insight into the mechanisms that underlie focus interpretation.

1.2 Previous accounts of ineffability

According to Schwarzschild [2004], the ineffability of (1) stems from a conflict between the requirements of free and associated focus.² On the one hand, focus in an answer to a wh-question is normally assigned to the constituent corresponding to the wh-phrase, which entails that *crepes* must be focused (free focus). On the other hand, *only* should associate with *Paris* to exclude alternative locations where John might eat crepes. This association can be ensured by focusing *Paris* (associated focus), thereby restricting the quantificational domain of *only* to the relevant set of alternatives.³

In each of the answers in (1), one of these demands is not satisfied: the focus on *crepes* in (1-a), while establishing the relation between question and answer, interferes with *only*'s association with *Paris* and leads to association with *crepes* instead, thereby producing the unwanted inference that crepes is the only food John eats in Paris. In contrast, in (1-b) the focus on *Paris* enables its association with *only*, but it does not signal that *crepes* is the answer to the question. Since *crepes* is not accented, it is treated as part of the background instead, which is incongruent with the preceding context.

The analysis is quite straightforward, but it cannot be the whole story, given that it is generally possible for *only* to associate with a constituent that does not bear the main accent. Consider (2), in which (2-b) (= (1-b)) serves as a response to (2-a), the mirror-image counterpart of the question in (1):

- (2) a. Where does John only eat CREPES?
- b. John only eats crepes in PARIS.

¹We indicate pitch accent using CAPS and take it to reflect an underlying F-feature marking focus. As we will see, however, F-marking is not always realized by pitch accent.

²The term *association with focus* refers to (truth-conditional) effects of focus in the scope of operators such as *only* and *even*, which are thus termed *focus-sensitive*. A *free focus* is focus that serves to establish discourse relations — marking contrast with a prior utterance, the answer to a previous question, or simply new information. Free focus is often contrasted with associated focus, which is viewed as “bound” by the focus-sensitive operator, by analogy with the distinction between free and bound pronouns.

³Assuming a simplified semantics for *only* as in (i), *only* takes a propositional argument p (its *prejacent*), presupposes that p is true, and asserts that all non-weaker alternatives (i.e., those not entailed by the prejacent) in the domain of quantification C are false.

(i) **Semantics for *only* (simplified):**
 $\llbracket \text{only} \rrbracket_C = \lambda p \lambda w : p(w). \forall q \in C [p \not\Rightarrow q \implies \neg q(w)]$

This formulation is simplified because blindly negating all non-weaker alternatives derives contradictions in some cases. A more refined approach holds that *only* negates just the *innocently excludable* (IE) alternatives — those that can all be jointly negated consistently with the assertion of the prejacent, without making arbitrary choices between alternatives (cf. [Fox, 2007]). For the present purposes, we may continue to use the simplified version, as our discussion does not hinge on this distinction. We will also ignore the distinction between assertion and presupposition for the most part.

Here, *crepes* and *in Paris* switch roles: *in Paris* is now the free focus corresponding to the wh-phrase in the question, while *crepes* serves as *only*'s associate. *Only* is apparently able to associate with *crepes*, even though the latter is not accented. The accent on *Paris* does not seem to interfere with this association. The demands of free and associated focus are just as much in conflict as in (1), yet the answer is perfectly acceptable.

Association with a prosodically reduced element is possible only when that same element has already occurred, associated with the same operator, in a previous utterance where it did bear accent. The associate in such cases is commonly referred to as a *second occurrence focus* (SOF) [Rooth, 1996, Féry and Ishihara, 2009].⁴ In (2-b), *crepes* is an SOF, having already appeared in the preceding question associated with *only*. The remaining puzzle is why *Paris* cannot receive a similar SOF interpretation in (1).

Examples involving SOFs have led Büring [2015] to propose that the source of the problem in (1) lies in the specific structural configuration. Rather than attributing ineffability specifically to a conflict between free and associated focus, Büring analyzes it as arising from a more general conflict between multiple foci. In fact, on his account free focus is just a special case of associated focus: every sentence contains a silent, root-level focus-sensitive operator that evaluates the free (discourse-related) foci in the sentence. So free focus is associated too — only with a covert operator rather than with an overt one.

Büring argues that ineffability arises from crossing dependencies between focus-sensitive operators and their associates. Schematically, the following configuration is argued to be ruled out by the grammar:

- (3) **Problematic configuration with crossing dependencies:**
 $*[Op_1 \dots [Op_2 \dots [F_1 \dots F_2]]]$

In discussing (1), Schwarzschild [2020] identifies Büring's free-focus operator with EXH [Fox, 2007, Chierchia et al., 2012], a covert exhaustification operator akin to overt *only*, which is argued to associate with the focus on *crepes*. The contribution of EXH — an exhaustivity inference — is intuitively confirmed by the possibility of paraphrasing the intended answer meaning with a cleft:

- (4) It's CREPES that John only eats in Paris.

The assumed structure for (1) is thus the following (subscripts serve to indicate focus association):

- (5) EXH₁ John only₂ eats [crepes]_{F1} [in Paris]_{F2}

In (5), dependencies between EXH, *only*, and their respective associates cross, in violation of (3) above. The situation in (2), the variant of (1) involving the mirror-imaged question, is different: here, dependencies between the two nested operators do not cross.⁵

⁴There is some evidence that SOFs are indeed foci, despite lacking the usual pitch accent — they exhibit prosodic marking through increased intensity and duration, and elements that resist prosodic prominence, such as reduced pronouns, cannot function as SOFs [Hoeksema and Zwarts, 1991, von Stechow, 1994, 1996, Beaver and Clark, 2003].

⁵The claim that the dependencies in (2) do not cross holds at least in terms of linear order, which is the primary concern in Büring [2015] (see Section 4.1). Although (2) appears structurally similar to (1), we argue in Section 3.1 that the two also differ in their underlying structure.

- (6) EXH₁ John only₂ eats [crepes]_{F2} [in Paris]_{F1}

Why should crossing dependencies between focus operators be problematic? Several possible explanations can be entertained. Büring [2015] derives (3) directly from general prosodic constraints; we review his account in Section 4.1 below. Schwarzschild [2020] views (3) as an intervention configuration, providing the following generalization:

- (7) **Schwarzschild’s generalization:** A focus-sensitive operator cannot associate across an intervening focus.

In (1), *only* should associate with *Paris*, but the focus on *crepes* gets in the way and blocks association. Similar intervention effects have been analyzed in terms of syntactic constraints on LF dependencies (cf. Beck [1996]), or as resulting from particular assumptions concerning the compositional system [Beck, 2006].⁶

We return to these possibilities and discuss them in more detail in Section 4.1 below. In what follows, we argue that there is no need to invoke any special mechanism: the problem with examples that involve crossing dependencies follows naturally from their semantic content, given the general requirement that they be interpreted as good answers to good questions. Essentially, we retain the core insight of Schwarzschild [2004] that the answer in (1) fails to relate appropriately to its preceding context. However, we argue that the relation between questions and answers cannot be reduced to a formal constraint on accent placement; it necessarily requires consideration of the answer’s semantic content and how it relates pragmatically to the question.

Our solution will not only be less stipulative than previous approaches; it will also enable us to account for a range of examples that appear to violate the constraint in (3). To preview the argument, consider the following example, due to Büring [2015]:

- (8) a. I only have a CREDIT card.
b. That’s quite alright, we only ACCEPT credit cards.

In (8-b), *only* appears to associate with *credit cards* across the intervening *accept*, which is unpredicted given formal approaches such as Beck [2006] or Büring [2015]. By contrast, the question-based perspective provides a natural explanation for the felicity of (8).

1.3 Overview

The remainder of the paper is organized as follows. In Section 2, we introduce a simple felicity condition on assertions: (a subset of) the set of focus alternatives to a sentence must correspond to the denotation of a good question, and the sentence must provide a good answer to that question. We then show that, regardless of how the answer in (1) is focused, it does not correspond to any good question in the

⁶Note that the generalization in (7) diverges from (3) in its empirical predictions. In particular, it rules out configurations with nested focus operators in which both foci lie within the scope of the embedded operator, regardless of whether the dependencies cross linearly. The acceptability of (2) is therefore attributed by Schwarzschild [2020] to structural differences between (1) and (2); see Section 3.1.1. For our purposes, however, the differences between (3) and (7) will not play a significant role. We will ultimately argue that even the more permissive constraint in (3) is overly restrictive; see Section 4.

given context. We characterize the problematic configuration as one that involves nested alternative computations, and requires an impossible covariation between alternative sets. The analysis will derive the difficulty posed by certain crossing focus dependencies, captured by the descriptive generalization in (3), directly from the interpretation.

In Section 3, we discuss central predictions of our analysis. We show that it predicts ineffability to arise only under specific structural conditions. In particular, we correctly predict that overt movement can obviate ineffability, while also accounting for the felicity of the mirror-image variant of the *crepes* example in (2). We then explore how the approach may extend to the well-known phenomenon of intervention effects in *wh*-in-situ questions, first identified by Beck [1996] and further studied by Beck [2006] and many others. We argue that intervention examples involve the same problematic semantic pattern that drives focus ineffability. We also show that unlike Beck’s approach, ours correctly predicts association with a focused *wh*-in-situ to be possible (cf. Li and Law [2016]).

In Section 4 we present an empirical argument against all purely formal accounts of ineffability, including those mentioned in Section 1.2 above. The argument is based on counterexamples to the formal constraint in (3), among them the *credit cards* example in (8). We take these examples to show that ineffability depends in part on contextual and other pragmatic factors. In particular, manipulating the question affects the felicity of the answer — which suggests that the question being addressed plays a crucial role in driving ineffability. Unlike previous accounts, the present one makes fine-grained predictions about the contextual conditions under which ineffability should arise, and is thus capable of distinguishing the problematic examples from felicitous ones that are formally very similar. Section 5 concludes.

2 Proposal

In this section, we propose an analysis of ineffability that derives the problem directly from interpretation — specifically, from the interaction between semantic content and general pragmatic principles — thereby eliminating the need for structural constraints like the one in (3). We argue that ineffability is, in fact, predicted under the view that focus signals the question addressed by an assertive sentence. In particular, we show that in (1), *only*’s domain of quantification cannot be established in a way that would make the answer congruent with the contextually supplied question, regardless of how focus is assigned. We argue that this incongruence results in the observed infelicity.

In Subsection 2.1, we lay out the general assumptions and principles upon which the analysis is built. In Subsection 2.2, we show how ineffability naturally follows from these assumptions and principles. Subsection 2.3 traces the source of the problem to a single operator’s quantificational domain being shared across all members of an implicit question.

2.1 Framework

The present proposal builds on a question-based approach to discourse, according to which assertions are always interpreted as answers to questions.⁷ In practice, discourse often involves implicit questions. The main focus of an utterance, generally realized by pitch accent, serves to reconstruct and accommodate an implicit question under discussion, against which the utterance is interpreted [Roberts, 1996,

⁷ “Every statement that anybody ever makes is made in answer to a question” [Collingwood, 1940].

Katzir, 2024].

Since Hamblin [1973] and Karttunen [1977], a question is taken to denote a set of alternative propositions, understood as the set of its possible answers. Focus, too, gives rise to a set of alternatives [Rooth, 1985, 1992]. The *focus alternative set* of an assertion is the set of propositional meanings obtained by replacing the main focus within the sentence with alternative elements of the same semantic type.⁸

The analysis relies on two general pragmatic conditions.⁹ The first relates focus to questions:

- (9) **F-TO-QUESTION:** The implicit question reconstructed from a sentence must be a subset of its focus alternative set.

For example, the set of focus alternatives to *John eats CREPES in Paris* is {John eats x in Paris | $x \in \{\text{crepes, bananas, ...}\}$ }, which corresponds to the question *What does John eat in Paris?*. The second condition is a felicity requirement based on questions and answers:

- (10) **Q-A-FELICITY:** The felicity of a sentence depends on whether the implicit question reconstructed from it is a good question, and whether the sentence answers it well.

This immediately raises the issue of what qualifies as a good question or a good answer.¹⁰ For the purposes of Sections 1-3, we may adopt a standard congruence requirement in the spirit of Rooth [1992]:

- (11) **Q-A-CONGRUENCE:** In a question-answer pair, the question must be a subset of the answer's focus alternatives.

Given Q-A-CONGRUENCE, an implicit question reconstructed from focus is considered good if it matches the explicit background question. Likewise, an answer is good if it is relevant to the background question (see Lewis [1988]; see Benbaji-Elhadad and Doron [2025 manuscript, under review] among others for a recent discussion of relevance). This condition is clearly too simplistic, given that explicit questions can often be answered indirectly, via accommodation of an implicit question that differs from the explicit one. For this reason, we treat F-TO-QUESTION and Q-A-FELICITY, rather than Q-A-CONGRUENCE, as the operative conditions governing focus assignment and association. However, we postpone discussion of such question shifts until Section 4.

Together, F-TO-QUESTION and Q-A-FELICITY constrain focus placement: an appropriate focus marking must yield both a good implicit question and a good answer to that question. Some focus

⁸Following Rooth [1985, 1992], the set of focus alternatives associated with a sentence is commonly referred to as its *focus semantic value* (FSV). We adopt the more neutral term *focus alternative set*, since we do not assume that focus alternatives are necessarily derived compositionally.

⁹We adapt the particular formulation of these conditions from Katzir [2024], though some version of them is often assumed in the literature more broadly.

¹⁰Recent work by Fox [2016, 2019, 2020] clarifies several central properties that make questions and answers good. For instance, Fox [2019] proposes PARTITION-BY-EXHAUSTIFICATION: the alternatives that form the denotation of a question, once exhaustified, must partition the context set — the set of worlds consistent with the presuppositions of the discourse participants. A good answer should then identify a cell (or a union of cells) in this partition, reflecting the speaker's current epistemic state.

assignments are ruled out because they correspond to inappropriate questions; others are excluded because they render the utterance irrelevant to the very implicit question reconstructed from it.

We further assume that *only* must associate with a focused element within its scope. This is Tancredi [1990]’s Principle of Lexical Association, which gains support from evidence by Beaver and Clark [2003, 2008]. We do not attempt to derive this principle, and we remain agnostic with respect to the mechanism by which association is established; it may be semantic, as in Krifka [1991] and Rooth [1985, 1992], or pragmatic, as in Vallduví [1990], Dryer [1994], Roberts [1996] and Schwarzschild [1997]. Importantly, the associate may be something other than the main (question-related) focus; in that case, it is prosodically reduced (i.e., an SOF). Like focus placement, focus association is restricted by the pragmatic conditions introduced earlier: patterns of association that lead to the reconstruction of a bad question, or that yield a bad answer to the reconstructed question, are ruled out.

2.2 Derivation of ineffability

We are now in a position to analyze the *crepes* example, repeated below, in terms of questions and answers. We assume that *only*’s domain of quantification is a contextual variable, indicated here with C:

- (1) What does John *only*_C eat in PARIS?
 - a. #John *only*_C eats CREPES in Paris.
 - b. #John *only*_C eats crepes in PARIS.

In (1), the background semantic question is:

- (12) $Q_{bg} = \{ \text{John eats } x \text{ in Paris and does not eat } x \text{ anywhere else} \mid x \in \{ \text{crepes, bananas, ...} \} \}$

For now, we set aside how this question meaning is arrived at; we return to this issue in Section 3.1.2. Concentrating on the answer, we consider the different possibilities for focus assignment, as well as for establishing *only*’s domain. We show that none of these options yields an interpretation in which the answer properly addresses the background question in (12); doing so would require an impossible variation in the domain of *only* across focus alternatives. Instead, all available parses force the accommodation of a question completely unrelated to Q_{bg} , which in turn results in the observed oddness.

Consider first (1-a), where *crepes* bears the accent and is therefore interpreted as the main focus. The implicit question reconstructed from this focus is:

- (13) $Q_a = \{ \text{John } \textit{only}_C \text{ eats } x \text{ in Paris} \mid x \in \{ \text{crepes, bananas, ...} \} \}$

If this is the only focus in the sentence, then the only available interpretation is one in which *only* associates with *crepes* — that is, $C = \{ \text{John eats } x \text{ in Paris} \mid x \in \{ \text{crepes, bananas, ...} \} \}$. On this interpretation, Q_a becomes $\{ \text{John eats } x \text{ and nothing else in Paris} \mid x \in \{ \text{crepes, bananas, ...} \} \}$, which can be paraphrased as *What food x is such that, when in Paris, John eats only x ?* Although this is a perfectly reasonable question in general, it is entirely orthogonal to the background question Q_{bg} , which asks about foods that John eats in Paris and nowhere else. Consequently, Q_a would not be considered a good question in the given context, in violation of Q-A-FELICITY.

Suppose now that *Paris* is also focused, though prosodically reduced (i.e. an SOF). In that case,

only may in principle associate with any of the foci in its scope: *crepes*, *Paris*, or both.¹¹ We have already seen what happens when *only* associates with *crepes* alone. If, instead, *only* associates with *Paris*, then $C = \{\text{John eats crepes in } y \mid y \in \{\text{Paris, London, ...}\}\}$. The implicit question Q_a becomes $\{\text{John eats } x \text{ in Paris and does not eat crepes anywhere (else): } x \in \{\text{crepes, bananas, ...}\}\} = \{\text{John eats crepes in Paris and nowhere else, John eats bananas in Paris and does not eat crepes anywhere, ...}\}$. As in the previous case, the reconstructed question is clearly unrelated to the background question Q_{bg} . (This time, it is also predicted to be odd in most contexts: it is difficult to imagine a natural scenario in which discourse participants would seek to determine which member of this Q_a is true. However, we are mainly concerned with the relation to the contextually supplied question.)

The remaining option is for *only* to associate with both foci, meaning $C = \{\text{John eats } x \text{ in } y \mid x \in \{\text{crepes, bananas, ...}\}, y \in \{\text{Paris, London, ...}\}\}$. In that case, Q_a becomes $\{\text{John eats } x \text{ in Paris, eats nothing else in Paris, and eats nothing outside of Paris} \mid x \in \{\text{crepes, bananas, ...}\}\}$. Once again, the resulting implicit question is quite odd in general, and unrelated to the background question Q_{bg} in particular, and so Q-A-FELICITY is violated. Thus, there is no available parse of (1-a) that satisfies Q-A-FELICITY, and the answer is predicted to be odd. A detailed derivation of (one parse of) this example, along with derivations for several other key examples discussed below, is provided in Appendix A.

In (1-b), the main focus is *Paris*, so the implicit question is:

$$(14) \quad Q_b = \{\text{John only}_C \text{ eats crepes in } y \mid y \in \{\text{Paris, London, ...}\}\}$$

As before, *only* may associate with *crepes*, *Paris*, or both. This situation is entirely parallel to that in (1-a) above (the only difference being that the accented element is here an adjunct rather than an argument, a distinction which is irrelevant for the analysis). The reader can verify that here, too, no possible parse yields an implicit question that is good given the background Q_{bg} . Consequently, that answer, too, is predicted to be infelicitous.

2.3 The source of the problem: impossible covariation

Consider what would be required for Q-A-FELICITY to be satisfied in (1-a). Ideally, the implicit question reconstructed from focus, $\{\text{John only}_C \text{ eats } x \text{ in Paris} \mid x \in \{\text{crepes, bananas, ...}\}\}$, should match the background question $\{\text{John eats } x \text{ in Paris and does not eat } x \text{ anywhere else} \mid x \in \{\text{crepes, bananas, ...}\}\}$. For this to happen, each focus alternative of the form *John only eats x in Paris* would require *only*'s domain to contain just the alternatives $\{\text{John eats } x \text{ in } y \mid y \in \{\text{Paris, London, ...}\}\}$, with the same x that appears in that particular focus alternative. This situation is illustrated in (15).

(15) **Question:** What does John only eat in PARIS?

Desired (and underivable) implicit question denotation:

¹¹If, as argued by Rooth [1992], association with focus is unselective, then the only available option would be to associate with both foci (although contextual pruning may still mimic selective association, even in the absence of a formal selective mechanism; see Erlewine [2025]). We will show that, whether association is selective or not, the unattested meaning cannot be derived.

{John only C_1 eats CREPES in Paris, John only C_2 eats BANANAS in Paris, John only C_3 eats CAKE in Paris}

$C_1 = \{\text{John eats crepes in Paris,}$
 John eats crepes in London,
 John eats crepes in Berlin}

$C_2 = \{\text{John eats bananas in Paris,}$
 John eats bananas in London,
 John eats bananas in Berlin}

$C_3 = \{\text{John eats cake in Paris,}$
 John eats cake in London,
 John eats cake in Berlin}

The problem, then, is that the alternatives forming *only*'s domain would have to covary with the main focus alternatives in order to allow for the reconstruction of a good implicit question. But this is impossible to derive: the alternatives in (15) may differ only in the value of the focused element. Since *only* is not focused, its domain must be assigned the same value across all focus alternatives (unless the domain C could somehow be focused by itself without simultaneously focusing *only*, which we assume is not possible).¹²

We can now offer a revised characterization of focus ineffability. Rather than attributing the problem to the mere presence of nested focus-sensitive operators, or to the particular pattern of crossing dependencies illustrated in (3), we take it to arise from (a) the need for covariation of alternative sets in order to satisfy the pragmatic purpose of the sentence (an answer to a particular kind of question), and (b) the impossibility of obtaining this covariation, regardless of whether we focus *crepes*, *in Paris*, or both.¹³

¹²That focusing the domain alone is not possible may follow from several considerations. One is that focus must be realized prosodically (even if not by pitch accent), which rules out focusing a silent element such as the domain variable. In addition, the domain might not be explicitly represented at LF, in which case it would not be available for F-marking in the first place. However, for the sake of clarity we will treat domain variables as syntactically represented.

In Section 4 we consider the possibility that *only* (not just its domain) is treated as focused, even though it is not accented. We argue that although this possibility is generally available, it turns out to be problematic in the *crepes* example (1). Our account relies on the assumption that focusing *only* does more than merely focus its domain.

¹³The *crepes* example can be replicated with other focus-sensitive operators, such as *also* and *even*:

- (i) What does John eat also in PARIS?
 - a. # John also eats CREPES in Paris.
 - b. # John also eats crepes in PARIS.
- (ii) What does John eat even in PARIS?
 - a. # John even eats CREPES in Paris.
 - b. # John even eats crepes in PARIS.

On our view, ineffability is predicted to arise when an operator's domain influences the implicit question reconstructed from focus in a particular way. Intuitively, *also* and *even* interact with the background question in a way parallel to *only*: for instance, the question in (i) roughly translates as *Which food x is such that John eats x in Paris, in addition to eating x somewhere else?* Congruence with this question requires *also*'s domain to vary with *x* across focus alternatives, which, we have argued, is impossible. However, a more explicit extension of the analysis to operators such as *also* and *even* depends on the semantics one adopts for these operators. In particular, the derivation may hinge on how presuppositions interact with the reconstruction of an implicit question — an interaction that is not yet fully understood.

A related issue is why ineffability does not arise even when the question lacks an explicit focus operator. Assuming that free focus is always associated with a covert EXH operator, one might expect the following pair to exhibit the same ineffability as in (1):

- (iii) a. What does John eat in PARIS?
- b. John only eats CREPES in Paris.

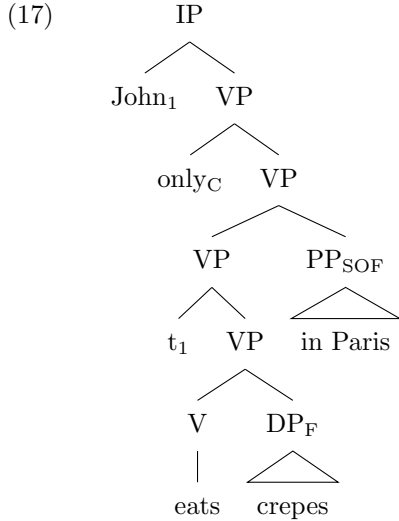
If *Paris* were associated with EXH in the question, (iii-a) would have been equivalent to the question in the original *crepes* example, *What does John only eat in PARIS?* We suggest that EXH is absent from this question; notably, there is no exhaustivity inference associated with *Paris* in (iii-a). This suggests that the tendency to interpret free focus in an assertion exhaustively does not automatically extend to questions.

Generalizing from our specific example, whenever the same operator appears in each member of an alternative set, its domain of quantification (itself an alternative set) cannot depend on the specific member of the larger set in which it occurs. In other words, the following situation is predicted to be problematic:

- (16) **Problematic constellation of nested alternatives:**
 $ALT = \{Op_C \dots x \dots | x\}$, where $C = \{x \dots y \dots | y\}$

Since x is fixed within C but varies across the larger alternative set, this larger set will contain alternatives in which the value of x inside C fails to match the value of the relevant part outside C . As we will see in Section 3.1, the interpretation can be rescued if x is a bound variable within C , since binding simulates such covariation.

The problematic situation in (16) is instantiated in the underlying structure of the *crepes* example (1), where the main (question-related) focus appears within the scope of a focus-sensitive operator that does not associate with it. The relevant LF is given in (17) (as discussed in Section 1.2, the example may involve a silent EXH operator; however, it can be ignored here, since it plays no role in our derivation of ineffability in (1)).¹⁴



The focused DP should vary across focus alternatives, yet remain constant within the operator's domain in each such alternative (since the operator does not associate with it). This yields the impossible requirement that the domain be assigned a different value in each alternative. The same problem does not arise when the focus-sensitive operator associates with the main focus that lies within its scope:

- (18) a. What does John eat in Paris?
b. John only eats CREPES in Paris.

¹⁴Following the VP-Internal Subject Hypothesis, we represent the subject as originating within the scope of *only*. We assume that it semantically reconstructs to influence the alternatives in the operator's domain. For expository clarity, we also annotate SOFs in the structure, though we take the relevant constituents to be simply F-marked.

Here, *only* associates with *crepes*, which is simultaneously interpreted as the main focus. In such cases, we correctly predict that no problem should arise. Although the operator’s domain $\{\text{John eats } x \text{ in Paris} \mid x \in \{\text{crepes, bananas, ...}\}\}$ is still shared across focus alternatives, in this case the operator can affirm or negate different domain members for each alternative:

- (19) **Focus alternatives:** {John only C eats CREPES in Paris,
 John only C eats BANANAS in Paris,
 John only C eats CAKE in Paris}
C = {John eats crepes in Paris, John eats bananas in Paris, John eats cake in Paris}

We believe that the present analysis is very much in line with Schwarzschild [1993]’s original discussion of the *crepes* example. Schwarzschild reduces question-answer congruence to a general contrast constraint on focus, requiring focus to mark a contrast with some contextually salient antecedent. He then shows that this constraint cannot be met by (1): if *only*’s domain is set up as the answer requires, the answer cannot contrast appropriately with other possible answers to the preceding question. Unlike Schwarzschild, we take questions and answers to be basic and treat contrast as derived from them (specifically, contrast holds between different answers to the same question). As we will see in Section 3.2, formulating the problem in terms of questions and answers allows us to make predictions about the interaction between focus and questions more generally — specifically, we predict so-called intervention effects in wh-questions [Beck, 1996, 2006].

3 Predictions

We have shown that ineffability follows from the alternative-sensitive computations taking place in the interpretation of focus and from how these computations feed the pragmatics of questions and answers. In this section, we consider two further predictions of the account. The first is that, if *crepes* in (1) is removed from the scope of *only*, the problem should be immediately solved, since *only*’s domain would no longer have to covary with focus alternatives. Thus, although we do not take the underlying structure itself to be the source of the problem, we can still explain why it matters, given that the resulting interpretation depends on the structure.

The second prediction is based on the assumption that wh-elements, like focused elements, function as sources of alternatives. If so, leaving the wh-phrase in (1) in situ should render the question itself infelicitous. This is because *only* appears in the question as well, and a wh-in-situ would recreate the problem of a question variable occurring inside *only*’s domain. In other words, the account correctly predicts problems to arise in wh-intervention constructions [Beck, 1996, 2006]. We summarize these two predictions below:

- (20) **Predictions made by the present account:**
1. Ineffability should not arise when the main focus lies outside the scope of the focus-sensitive operator.
 2. Intervention is predicted when a wh-element appears within the scope of a focus-sensitive operator.

As we will see, the second prediction has an important caveat — in case the operator associates with the wh-element itself, intervention is no longer predicted (and this, too, is borne out). We now discuss these predictions and their implications in turn.

3.1 Prediction 1: felicitous variants of (1)

There are several ways to remove the main focus (in our example, *crepes*) from the scope of *only*: the structure can be altered such that the main focus is not within *only*'s scope to begin with, or it can be moved out (for example, through topicalization or extraposition). The prediction is that either way should yield an acceptable answer. As we will see, this prediction is borne out.

3.1.1 Examples involving SOFs

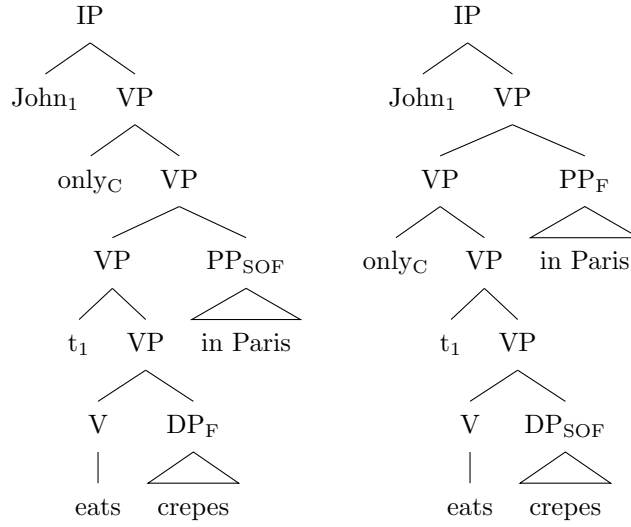
Consider again the variant of the *crepes* example in which the direct object *crepes* and the adjunct *in Paris* switch roles:

- (2) a. Where does John only eat CREPES?
- b. John only eats crepes in PARIS.

(2-b) is acceptable as an answer to (2-a). In this case, *only* associates with *crepes*, even though the latter is prosodically reduced; it is interpreted as a second occurrence focus (SOF), while *Paris* is interpreted as the main (question) focus. Apparently, the presence of the focus on *Paris* does not interfere with deriving *only*'s association with *crepes*.

We join Schwarzschild [2020] in viewing the contrast between (1) and (2) as reflecting a difference in underlying structure. Specifically, we propose that in (2), *in Paris* is in fact not in the scope of *only*, but is attached higher in the structure. This possibility is not available in the original *crepes* example, where *only* should associate with *Paris* and therefore the latter must be within the operator's scope. Compare the LF for the original example (1) (on the left) with its mirror-image variant in (2) (on the right):¹⁵

¹⁵This raises the question of why *crepes* cannot take scope above *only* via covert movement in the structure on the left. Such covert movement seems to be unavailable, an issue we will return to in Section 3.1.2.



In constructions parallel to (2) where the structure is fixed by variable binding, the example becomes degraded:

- (21) a. What picture did John show only to some MOTHER?
b. # John only showed some mother a picture of HER SON.

(21-b) cannot mean that John showed some mother a picture of her son and did not show that picture to anyone else. Rather, the only available reading is one where John did not show the mother a picture of anyone other than her son. In other words, association with *some mother* is impossible. The crucial difference from (2) is that in (21), *her son* is bound by *some mother*, and therefore it must be within *only*'s scope (as long as *some mother* is, which is required for it to be associated with *only*):

- (22) only [John showed [some mother]_i a picture of her_i son]

The problem with (21) can be derived in the same way that allowed us to account for ineffability in the original *crepes* example (1): once the main focus appears within *only*'s scope, it entails an impossible covariation of the operator's domain with question alternatives. In contrast, in (2) such covariation is not required, since the main focus no longer affects *only*'s domain in any way.¹⁶

¹⁶A similar explanation can be proposed for the felicity of a variant of (1) in which *only* appears in the answer adjacent to the PP, rather than the entire VP:

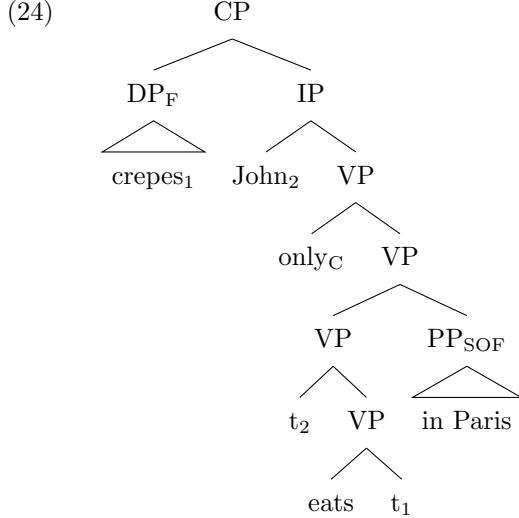
- (i) a. What does John only eat in PARIS?
b. John eats CREPES only in Paris.

Once the scope of *only* is restricted to the PP, *crepes* does not appear in the alternatives that form *only*'s domain, and the problematic ineffability configuration does not arise. This analysis requires *only* to undergo type-shifting so that it can operate over an argument that does not denote a proposition; [*only in Paris*] must then QR as a unit to compose with the verb. Initial evidence for such movement comes from examples involving antecedent-contained deletion (ACD) [Crnic, 2013]. However, Hirsch et al. [2017], Bassi et al. [2022] provide compelling arguments against composing *only* directly with a quantifier, and argue that it is always interpreted as a propositional operator at the VP-level. If this is correct, any account of ineffability that derives the problem at LF would have a hard time explaining the contrast

3.1.2 Movement obviates ineffability

As Schwarzschild observes, overt movement obviates focus ineffability: (23-b), with its corresponding LF given in (24), is a perfectly acceptable answer to (23-a).

- (23) a. What does John only eat in PARIS?
b. CREPES, John only eats in Paris.



Assume that *only* associates with *Paris*, that is, $C = \{\text{John eats } t_1 \text{ in } y \mid y \in \{\text{Paris, London, ...}\}\}$. Here, the trace functions as a bound variable. The interpretation of the IP states that John eats t_1 in Paris and does not eat t_1 anywhere else. After lambda abstraction and binding by *crepes*, this yields the correct meaning — namely, that John eats crepes only in Paris.

If focus alternatives are obtained by replacing focused constituents within a given LF with their alternatives (cf. Katzir [2007], Fox and Katzir [2011]), each alternative to *crepes* similarly binds into *only*'s domain in the relevant alternative. Covariation of *only*'s domain with the focus alternatives can then be simulated through the mechanism of variable binding, as in (25):

- (25) **Focus alternatives:** {CREPES₁ John only_C eats t_1 in Paris,
BANANAS₁ John only_C eats t_1 in Paris,
CAKE₁ John only_C eats t_1 in Paris}

$$C = \{\text{John eats } t_1 \text{ in Paris, John eats } t_1 \text{ in London, John eats } t_1 \text{ in Berlin}\}$$

Due to the domain being bound in each alternative, the implicit question reconstructed from focus becomes $\{\text{John eats } x \text{ in Paris and does not eat } x \text{ anywhere else} \mid x \in \{\text{crepes, bananas, ...}\}\}$. This question is identical to the background Q_{bg} in (12), so trivially, the question and answer are both

between *John eats CREPES only in Paris* and *John only eats CREPES in Paris*. The analysis in Buring [2015] appears to have an advantage in this regard, though Section 4 discusses other challenges it faces.

good given Q_{bg} , and Q-A-FELICITY is satisfied. Note that having the trace appear both within *only*'s domain and within its prejacent in each focus alternative is essential — the same strategy could not be applied to the original *crepes* example in (1), where no such trace is present in the prejacent.

We can now explain how the derivation of the background question itself, (23-a), follows through. It is parallel to the derivation of (23-b): the wh-trace in the scope of *only* is bound from the outside, so covariation of the domain with question alternatives is guaranteed, and no problem arises:¹⁷

(26) What_{*i*} does John only_C eat t_{*i*} in PARIS?

In general, whenever a variable that appears in an embedded alternative set is bound from the outside, covariation between the inner and outer alternatives becomes possible. This also explains why ineffability does not arise in simple cases such as the following:

(27) [Every boy]_{*i*} thinks that Mary only saw him_{*i*} in PARIS.

In this example, the pronoun is bound within the scope of *only*, allowing the operator's domain to range over boys along with the universal quantifier, such that for each boy x , *only*'s domain is $\{\text{Mary saw } x \text{ in } y \mid y \in \{\text{Paris, London, ...}\}\}$.

A natural question at this point is why covert movement of *crepes* — parallel to the overt movement we have just considered — cannot rescue the original example in (1) by simulating the required covariation of alternatives. As a DP, *crepes* should in principle be able to QR above *only*, thereby avoiding the problematic configuration. We must therefore assume that covert movement is impossible in this example.¹⁸ Note that the possibility of covert movement poses a challenge for any account that derives ineffability from constraints on LF, including Schwarzschild [2020] and the extension of Beck [2006] we will consider in Section 4.1 below.¹⁹

The question of why QR does not apply in this example remains an open one. However, we would like to highlight some independent evidence suggesting that such covert movement may in fact be blocked. In general, adverbial (pre-VP) *only* appears to “freeze” the scope of focused quantifiers within its syntactic domain. Consider the following example from Aoun and Li [1993]:

(28) a. Someone loves every boy in the room. ($\exists \gg \forall, \forall \gg \exists$)
b. Someone only loves [every BOY]_F in the room. ($\exists \gg \forall, *\forall \gg \exists$)

¹⁷Pointwise Functional Application — the semantic composition rule in the frameworks of Hamblin [1973] and Rooth [1992] — has been argued to be incompatible with Predicate Abstraction and variable binding [Shan, 2004]. For the example in (26), we assume that lambda abstraction applies within each member of the (here singleton) Hamblin set that combines with the binder index. Developing a more principled framework that integrates Hamblin semantics with variable binding remains an open problem. Note that any theory of focus intervention that adopts a Hamblin-style semantics for wh-phrases (including Beck [2006], Li and Law [2016] and Erlewine [2025]) faces similar challenges. For a proposal that attempts to derive intervention effects from this very issue, see Kotek [2017].

¹⁸We do not rule out the possibility that for some speakers, covert movement is nonetheless employed as a kind of repair strategy (see Ross [1969], Lasnik [2001], Fox and Lasnik [2003] for discussion of repair strategies in the context of wh-island violations). This possibility might account for some variation in acceptability judgments for the *crepes* example in (1).

¹⁹The analysis in Büring [2015], being grounded in prosodic realization, has the potential advantage of avoiding this challenge, but see Section 4 for challenges for Büring's account.

This effect has been attributed by Aoun and Li [1993] and Erlewine [2014] to *only*'s inability to associate with extracted elements [Tancredi, 1990, Beaver and Clark, 2008]. On this view, *every boy* is prevented from undergoing QR because it is the associate of *only*. We would like to entertain an alternative explanation according to which it is the focus on *every boy* that is responsible for the scope-freezing effect. Indeed, the inverse scope in (28-a) is far less accessible when *boy* is narrowly focused, even in the absence of *only*.

Given this observation, if *crepes* in (1) is focused, its covert movement should indeed be blocked. If, on the other hand, *crepes* is not focused, QR may be possible. Even so, the result (crepes_i , John only eats t_i in PARIS) would still fail to invoke an implicit question matching the background, since question alternatives would be derived from the focus on *Paris*, not from *crepes*.

3.1.3 Interim summary

We have shown that a theory based on the minimal assumption that utterances are interpreted as answers to questions can derive ineffability in the *crepes* example. The problem arises because a question-related alternative source — the main focus — occurs within the scope of *only* without being its associate. This leads to conflicting requirements regarding the domain of *only*: each member of the question denotation (that is, each focus alternative to the answer) requires a different domain for *only*. However, the grammar does not allow the domain to vary across alternatives. In this way, the theory derives the limited distribution of nested foci directly from interpretive principles, without invoking formal stipulations or ad hoc constraints that categorically prohibit crossing dependencies between focus-sensitive operators.

We have also considered the different ways of removing the problematic alternative source from the scope of *only*: it can be overtly moved out, or it may simply not be in *only*'s scope from the start (as in (2), the mirror-image SOF variant of the *crepes* example). Covert movement of *crepes* alone is apparently not a possibility, potentially due to a scope-freezing effect of focus. We now turn to our second prediction, that if the wh-element in (1) were left in situ, the interpretation of the question itself should become problematic — in other words, intervention effects are predicted.

3.2 Prediction 2: intervention effects

3.2.1 Ineffability and intervention

A general corollary of the discussion above is that whenever the same domain of quantification appears inside each member of a larger alternative set, ineffability might arise. We therefore expect problems to arise when a focus-sensitive operator occurs inside a wh-question — which itself denotes a set of alternatives — if the wh-element lies within the operator's scope. This prediction is borne out in the form of the well-known intervention effects found in questions with wh-in-situ in many languages, including Mandarin [Li and Law, 2016]:

- (29) *Ta zhi yaoqing-le LIBAI chuxi shenme huodong? (Mandarin)
 he only invite-Asp Libai attend what activity
 Intended: 'What was the activity x such that he invited only Libai to attend x?'

In the relevant cases, an intervening operator — in this example *zhi*, the Mandarin counterpart of English *only* — appears to interfere with the interpretation of the wh-in-situ. Beck [2006] observes that intervention has to do with focus interpretation: cross-linguistically, the main interveners are

the focus-sensitive *only*, *even*, and *also*. She takes the interpretation of a wh-in-situ to depend on a question operator Q , and identifies the following schematic structure as problematic:

- (30) **Wh-intervention configuration:**
 $*[Q \dots [Op \dots [XP_F \dots wh]]]$

We discuss Beck’s proposal and its shortcomings in more detail in Section 4.1. For now, we note the similarity between Beck’s characterization of an intervention configuration and the configuration identified by Buring for cases of ineffability, repeated below:

- (3) $*[Op_1 \dots [Op_2 \dots [F_1 \dots F_2]]]$

(30), too, is an instance of crossing dependencies between operators and the elements associated with them, only this time one of these operators is the question operator Q , and its “associate” is the wh-in-situ.

Another striking parallel between ineffability and intervention is that, in both cases, overt movement can prevent the grammatical breakdown. We have already seen how overt movement can obviate ineffability; as Beck observes, fronting the wh-element likewise obviates intervention:

- (31) Shenme huodong, ta zhi yaoqing-le LIBAI chuxi? (Mandarin)
 what activity he only invite-Asp Libai attend
 ‘What was the activity x such that he invited only Libai to attend x?’

Given the similarities just noted, it appears plausible that the breakdown observed in intervention and ineffability stems from a common underlying mechanism. We will now show that, under a Hamblin semantics for wh-in-situ, intervention constructions are indeed predicted to be problematic, in a way that closely resembles our explanation of ineffability above.

3.2.2 Intervention and impossible covariation of alternatives

Following Hamblin [1973], we assume that wh-phrases denote sets of alternative individuals — for example, *who* denotes the set of all (relevant) human individuals, and *what activity* — the set of (relevant) activities. These alternatives are expanded compositionally via Pointwise Functional Application, which combines a set of functions with a set of arguments by applying each function to each argument. In this way, wh-phrases can be interpreted in situ and acquire scope within the semantics, without movement (this view stands in contrast to approaches that posit covert wh-movement, such as Karttunen [1977], Heim [2000]).²⁰

Given these assumptions, let us compare the intervention example (29) with its grammatical counterpart (31), in which the wh-element has been fronted:

²⁰We do not claim that an in-situ interpretation for wh-in-situ is *a priori* superior to a movement-based approach. Nevertheless, under a Hamblin-style semantics we can link wh-intervention to focus ineffability — a connection that appears difficult to establish using Karttunen semantics, especially since there are strong reasons to assume that focus itself is interpreted in situ (see e.g. Bassi and Longenbaugh [2020]).

- (29) *Ta zhi yaoqing-le LIBAI chuxi shenme huodong? (Mandarin)
 he only invite-Asp Libai attend what activity
 Intended: ‘What was the activity x such that he invited only Libai to attend x ?’
- (31) Shenme huodong, ta zhi yaoqing-le LIBAI chuxi? (Mandarin)
 what activity he only invite-Asp Libai attend
 ‘What was the activity x such that he invited only Libai to attend x ?’

The relevant question meaning requires *zhi* to exclude, for each activity x , the invitation of other individuals besides Libai to x . In other words, *only*’s domain should depend on the specific question denotation member in which it occurs. In (31), this covariance between domain and question alternatives is achieved through wh-movement. The derivation is entirely parallel to the cases involving overt movement discussed in Section 3.1.2 above: the domain of *only* contains the trace of the wh-phrase — a bound variable. Lambda abstraction above *only* ensures that, in each question alternative, the trace is bound by the corresponding wh-alternative, allowing the domain to depend on the particular question alternative in which it occurs.²¹

A parallel derivation is not available for (29). The alternatives generated by the wh-element give rise to the question denotation {he only_C invited Libai to attend x | $x \in \{\text{the dinner, the ball, ...}\}$ }. In each member of this set, that is for each x , *only* should negate alternatives of the form {he invited y to attend x | $y \in \{\text{Libai, John, Mary, ...}\}$ }. However, without movement and binding, covariation of *only*’s domain with the question alternatives is impossible.

Just as in the *crepes* example, we must assume that covert wh-movement is not possible here — if it were, it would resolve the problem. We do not know why such movement is blocked; perhaps this freezing effect is once again caused by the presence of a focus-sensitive operator. Importantly, this challenge is not unique to our account: in fact, any analysis of intervention effects that interprets wh-phrases in situ, including Beck [2006], Li and Law [2016] and Erlewine [2025], must likewise explain why covert movement of the wh-phrase is unavailable in the relevant examples.²²

We do not aim to provide a comprehensive account of intervention effects here. Rather, we focus on their connection to focus ineffability and show that the same line of reasoning which allowed us to derive the latter also predicts wh-intervention constructions to be problematic. In particular, we account for the fact that (29) cannot receive the interpretation that is obtained by fronting the wh-phrase. A fully developed account of intervention must also rule out other conceivable interpretations, given that the example is judged ungrammatical. We suspect that the exclusion of these alternative interpretations may likewise be motivated by considerations of questions and answers. However, pursuing this issue lies beyond the scope of the present paper.²³

²¹A complication arises when movement is combined with Hamblin semantics, though the simple cases considered here do not pose a serious problem. See footnote 17.

²²A possible explanation is suggested by Beck [1996]. Although intervention effects occur in many typologically unrelated languages, there is some cross-linguistic variation. For example, in English, intervention is more restricted. Beck observes that intervention effects tend to occur in languages that permit overt scrambling. In such languages, scope relations can be made explicit at surface structure, and Beck hypothesizes that they therefore must be. If covert movement is more freely available in languages that lack overt scrambling, such as English, intervention is predicted to be obviated in these languages.

²³There may also be independent compositional or pragmatic grounds for excluding some of these possible interpretations; for two related proposals, see Li and Law [2016] and Erlewine [2025].

3.2.3 A note on F-WH association

One of the main challenges to the account of intervention proposed by Beck [2006] comes from examples in which a focus-sensitive operator associates with a wh-in-situ, when that wh-phrase is focused. Consider the following Mandarin example from Li and Law [2016]:

- (32) Libai zhi chuxi-le SHENME HUODONG? (Mandarin)
 Libai only attended what activity
 ‘What was the activity x such that Libai only attended x ?’

Li and Law present evidence that in this example, *zhi* does in fact associate with the wh-phrase. As Li and Law observe, this phenomenon is entirely unexpected under Beck’s framework (see Section 4.1 below). By contrast, on the present account F-WH association follows directly, with no further assumptions: the denotation of (32) is $\{\text{Libai only}_C \text{ attended } x \mid x \in \{\text{the dinner, the ball, ...}\}\}$. Setting the operator’s domain to $C = \{\text{Libai attended } x \mid x \in \{\text{the dinner, the ball, ...}\}\}$, we get the overall question meaning $\{\text{Libai attended } x \text{ and no other activity} \mid x \in \{\text{the dinner, the ball, ...}\}\}$. This is precisely the attested meaning; we therefore correctly predict the example to be felicitous.²⁴

Notice that F-WH association is in fact a special case of a more general situation, in which an operator associates with a question-related alternative source within its scope — be it the wh-element in a question, or the main focus in an assertion (i.e., this is the typical situation whenever an operator’s associate bears the main accent). In that case, the operator’s domain alternatives and the question alternatives coincide, so their covariation is guaranteed. Therefore, on the preset view F-WH association is unsurprising.

²⁴One might ask whether the possibility of F-WH association undermines our explanation of wh-intervention. Could the focus-sensitive operator likewise associate with the wh-in-situ to rescue the intervention construction in (29)? Recall that (29) denotes the set $\{\text{he only}_C \text{ invited Libai to attend } x \mid x \in \{\text{the dinner, the ball, ...}\}\}$. Setting $C = \{\text{he invited Libai to attend } x \mid x \in \{\text{the dinner, the ball, ...}\}\}$ yields the question $\{\text{he invited Libai to attend } x \text{ and no other activity} \mid x \in \{\text{the dinner, the ball, ...}\}\}$, which is, in general, a good question (albeit not the one reported as intended).

We argue that this parse is nonetheless problematic, due to the focus on *Libai* in (29). Notice that association with the wh-in-situ with nothing else going on in the background would leave this focus unjustified (this is unlike the ineffability example (1), where the main focus is independently licensed by the background question). Consequently, such association should only be possible if there is independent justification for the focus on *Libai*. Suppose that, by analogy with the way focus in an assertion introduces alternative propositions, focus within a question introduces alternative questions — questions that are identical to the original one, except that the focused element is replaced by its alternatives. In (29), focus on *Libai* gives rise to alternative questions of the following form:

- (i) a. $\{\text{he only}_C \text{ invited Libai to attend } x \mid x \in \{\text{the dinner, the ball, ...}\}\}$
 b. $\{\text{he only}_C \text{ invited John to attend } x \mid x \in \{\text{the dinner, the ball, ...}\}\}$
 c. $\{\text{he only}_C \text{ invited Mary to attend } x \mid x \in \{\text{the dinner, the ball, ...}\}\}$

Assuming that *only*’s domain must be the same across all of these questions, and given our choice of value for C above, every alternative in that domain will make reference to *Libai*. For example, we get $\{\text{he only}_C \text{ invited Mary to attend } x \mid x \in \{\text{the dinner, the ball, ...}\}\} = \{\text{he invited Mary to attend } x \text{ and he did not invite Libai to attend any activity} \mid x \in \{\text{the dinner, the ball, ...}\}\}$. Thus, even though the original question is a good one, the focus within it generates a family of other, odd questions. In contrast, when the wh-phrase itself bears the main focus, as in (32), the same issue does not arise.

3.2.4 Interim summary

In this section, we have discussed two major predictions of the present account: first, by removing the main focus from *only*'s scope in (1), ineffability is obviated; and second, when a wh-in-situ appears within the scope of a focus-sensitive operator, intervention arises. Ineffability and intervention share the following properties: in both cases, a question-generating alternative source (*crepes* in (1), the wh-in-situ in (29)) lies within the scope of a focus-sensitive operator (*only* and *zhi*, respectively). Crucially, the operator is intended to associate with something else within its scope (*Paris* in (1), *Libai* in (29)). A schematic representation of this situation is given below:

$$(33) \quad *[\text{Question} \dots [\text{Focus operator} \dots [\text{Question-related alternative source} \dots \text{Focus associate}]]]$$

We have derived the problem with this configuration from the impossibility of covariation between the focus operator's domain and the question alternatives. In other words, the structure itself is not inherently problematic; rather, it produces a problematic interpretation — at least, not the interpretation one would expect.

4 A challenge for purely formal accounts of ineffability

This section presents a general challenge for theories that derive ineffability from structural configuration: namely, the existence of apparent counterexamples, in which a similar configuration does not result in ineffability. On the basis of such cases, Rooth [2010] and Beaver and Velleman [2011] conclude that ineffability is theoretically uninformative; crossing dependencies between focus operators are taken to be generally permitted, and the difficulty of the *crepes* example remains unexplained. We instead take these counterexamples as evidence that ineffability depends, in part, on pragmatic considerations.

Subsection 4.1 reviews previous approaches that formally derive the ban on crossing dependencies from Section 1.2 above. Subsection 4.2 examines felicitous cases where an accented verb intervenes between an operator and its associate; crucially, in these examples the question being addressed is not fixed by the context. Because our analysis concerns question-answer pairs rather than the answer in isolation, we explain these contrasts, which might otherwise seem puzzling. Finally, Subsection 4.3 discusses the possibility of accommodating an implicit question that differs from the contextually-supplied explicit one. This subsection highlights the pragmatic component of our account, showing how factors such as world knowledge and general reasoning are expected to influence the felicity of examples like (1).

4.1 Formal approaches to ineffability

Unlike the present account, which focuses on how semantic content feeds the pragmatic question-answer relation, previous analyses of ineffability trace the problem to a particular structural configuration instantiated in the answer. Recall the configuration identified by Büring [2015]:

$$(3) \quad \textbf{Problematic configuration with crossing dependencies:} \\ *[\text{Op}_1 \dots [\text{Op}_2 \dots [\text{F}_1 \dots \text{F}_2]]]$$

Schwarzschild [2020] hints at the possibility of deriving this generalization from compositional semantics by adopting the proposal in Beck [2006, 2016]. Beck is primarily interested in the wh-intervention

constructions discussed in Section 3.2; however, her general framework predicts intervention effects to arise from crossing dependencies between multiple focus-sensitive operators too.

Following Rooth [1985, 1992], Beck assumes that focus alternatives are propagated compositionally in a separate semantic dimension. Association with focus results from an operator’s domain of quantification being restricted to a subset of its complement’s focus alternatives. Crucially, a focus-sensitive operator must associate unselectively with all foci in its scope. This alone is sufficient to derive Schwarzschild’s generalization in (7) that an operator cannot associate across an intervening focus: the operator simply cannot ignore a focus that lies within its scope.

Beck further assumes that focus association “uses up” alternatives and prevents them from propagating further, by resetting the value in the alternative semantic dimension to the ordinary interpretation. This means that association with one operator renders focus invisible to higher operators. Recall the underlying structure assumed for (1):

$$(5) \quad [\text{EXH}_1 \text{ } [_{\text{TP}} \text{ John only}_2 \text{ } [_{\text{VP}} \text{ eats } [\text{crepes}]_{\text{F1}} \text{ } [\text{in Paris}]_{\text{F2}}]]]$$

In this structure, the lower operator *only* intervenes between the covert EXH and its associate, *crepes*. Therefore, EXH cannot access the alternatives generated by *crepes*, which have already been “used up” by *only*. Intervention effects are therefore predicted to arise. In other words, Beck predicts ineffability to be an instance of intervention.

Despite its initial appeal, Beck’s account is problematic both on conceptual and on empirical grounds. Her derivation of intervention is highly stipulative, relying crucially on the postulation of an additional semantic dimension, and on the assumptions that focus association is unselective and resets alternatives. The analysis of wh-intervention requires additional stipulations: question alternatives and focus alternatives must be the same semantic objects, occupying the same alternative semantic dimension; it is further assumed that wh-elements lack an ordinary semantic value and that their interpretation relies on a designated question operator *Q*. We have already seen that the present account requires no such stipulations; it is therefore preferable on grounds of parsimony.

Beyond this conceptual concern, Beck’s account also faces serious empirical challenges, most notably due to the assumption that focus association resets alternatives. As noted by Li and Law [2016] and Erlewine [2025], Beck’s system incorrectly predicts intervention effects whenever multiple operators associate with the same focused element: the lower operator would reset focus alternatives and block association with the higher one. Given the assumption that a wh-in-situ must be associated with a question operator *Q*, the same issue arises for cases of F–WH association, in which the wh-in-situ also associates with a focus-sensitive operator. In Section 3.2.3, we showed that such patterns are straightforwardly derived under the present account: once the operator’s associate is the wh-element, variation of the operator’s domain across question alternatives is guaranteed. More generally, multiple association patterns are correctly predicted to be unproblematic, because when two operators associate with the same element, the lower operator’s domain automatically covaries with the higher operator’s domain. See Appendix A for an example derivation.²⁵

²⁵Another issue for Beck stems from the fact that reducing wh-alternatives to focus alternatives wrongly predicts a focus within a question to be interpreted as a wh-element, a problem already noted by von Stechow [1991] and further discussed by Erlewine [2025]. Beck’s proposed solution is that, unlike other focus-sensitive operators, *Q* is selective: it evaluates only the wh-elements with which it is coindexed, rather than all alternative sources in its scope [Beck, 2006]. This distinction between *Q* and other operators is rather stipulative, and unexpected if both are assumed to rely on the same semantic mechanism.

Büring [2015] offers an alternative to Beck’s semantic approach. He derives the problem with the crossing dependencies configuration in (3) from the following constraint on prosodic realization:

- (34) **FOCUS PROMINENCE:** The metrically strongest (most prominent) element in the prosodic domain²⁶ of a focus-sensitive operator *O* (whether overt or covert) must be a focus associated with *O*.

In (5), it is impossible to satisfy FOCUS PROMINENCE simultaneously for both EXH and *only*, since each of these operators’ domain contains both foci. If *crepes* is more prominent than *Paris*, FOCUS PROMINENCE is violated for *only*. Conversely, if *Paris* is more prominent, FOCUS PROMINENCE is violated for EXH.

Büring’s approach is less stipulative than Beck’s, as it derives ineffability from a single, general principle (although, as we will see in Section 4.2, he does appeal to additional stipulations in an attempt to account for certain problematic cases). In addition, Büring’s proposal correctly allows for multiple associations with the same focus. We note that it is unclear whether this account could be extended to wh-intervention: since a wh-element generally does not need to be the most prominent element in a question, competition for prosodic prominence is unlikely to arise between a focus and a wh-in-situ. Consequently, intervention effects cannot be derived from FOCUS PROMINENCE in the same way that ineffability can. A more pressing issue, however, is that FOCUS PROMINENCE appears to be empirically inadequate; we turn to this issue now.

4.2 Ineffability depends on the question

The literature discusses cases in which an intervening accent apparently does not interfere with focus association. Such cases include (8), repeated from the introduction, and other similar examples:

- (8) a. I only have a CREDIT card. [Büring, 2015]
b. That’s quite alright, we only ACCEPT credit cards.
- (35) People who GROW rice only EAT rice. [Rooth, 1992]
- (36) a. Radiology? Why did you take radiology? [Schwarzschild, 2020]
b. They only OFFERED radiology.

In these examples, the focus-sensitive operator *only* is able to associate with the direct object across an intervening accent: (8), for instance, conveys that the speaker accepts credit cards and nothing else, not that they engage in no other actions involving credit cards besides accepting them. On the surface,

²⁶Büring defines the prosodic domain of a focus as follows:

- (i) **Prosodic focus domains:**
- a. A *p-domain* is a continuous sequence of prosodic phrases (e.g., prosodic words) contained within a single prosodic phrase at the next higher level in the prosodic hierarchy.
 - b. The *prosodic focus domain* of a focus-sensitive operator *O* and its associated focus *F* is the biggest *p-domain* that contains *F* but excludes *O*.

The prosodic domain of a focus roughly corresponds to the syntactic scope of the operator with which it associates, though as we will see in Section 4.2, there may be exceptions to this correspondence.

this pattern is reminiscent of the ineffability configuration. Assuming that the accented verbs function as free foci, the examples exhibit crossing dependencies between a free-focus operator, *only*, and their respective associates; yet, no ineffability arises. Such examples thus pose a serious challenge both for Beck [2006] and for Büring [2015]: they arguably demonstrate that focus association must be selective, since in all relevant cases the focus-sensitive *only* does not associate with the accented element that lies within its scope; the examples further appear to violate Büring’s FOCUS PROMINENCE constraint, since in all of them *only* is able to associate with a prosodically reduced element across an intervening accent.

In an attempt to account for such examples, Büring argues that in principle, there can be material between an operator and its associated focus which does not belong to the operator’s prosodic focus domain. Such material can be more prominent than the focus without violating FOCUS PROMINENCE, and by stipulation, this material is limited to a single prosodic word. The contrast between the ineffable *crepes* example (1) and the felicitous examples (8), (35)-(36) is thus attributed to differences in prosodic complexity: in the *crepes* case, *eats crepes* does not form a single prosodic word, whereas in the *credit cards* example, for instance, *accept* does.

However, we point out that this explanation cannot be right, since we can easily complicate the example without affecting its felicity:

- (37) a. I only have a CREDIT card.
 b. That’s quite alright, we’re only allowed to ACCEPT credit cards.

Büring’s account therefore appears too inflexible to capture the relevant contrasts. In contrast, the present account is question-based and in fact it treats ineffability as a property of a specific question-answer pair rather than of the answer in isolation. We can therefore account for the felicity of (8), (35) and (36) by appealing to contextual differences relative to the problematic (1), in particular to differences in the questions these examples address.

Since the context in (8) and (35) does not provide an explicit question, we take these examples to address an implicit one, which is inferred from their focus structure. We propose that the questions addressed by these examples are in fact not parallel to the question in (1). Specifically, we take the accent in these examples to reflect a broader focus, which also includes the operator *only*. This idea gains some support from the fact that, in the relevant examples, *only* can indeed bear the main accent. For example:

- (38) a. I only have a CREDIT card.
 b. That’s quite alright, we ONLY accept credit cards.

Under the assumption that focusing *only* results in the corresponding sentence without *only* being an available alternative, we may treat *only* in (8) and (35) as omitted from the implicit question reconstructed from focus.²⁷ As for (36), it may also involve the accommodation of an implicit question, or we may simply take (36-b) to directly answer the explicit question in (36-a). Either way, *only* can reasonably be assumed to be left out of the question in this case as well.

²⁷ *Only* has been argued by Dryer [1994] to be interpreted as part of the focus quite often. Without this possibility, many question-answer pairs in which a focus-sensitive operator appears in the answer, but not in the question, would be incongruent. One such example is discussed by Schwarzschild [1993]:

(i) a. Who did John see at the conference?

Once *only* does not appear in the implicit question, it no longer influences the interpretation of that question. In particular, its quantificational domain need not covary with the question alternatives in order for that question to be good. The question reconstructed from the broad focus is thus a good one, and the target sentence answers it well. In short, in examples (8) and (35)-(36), either there is no explicit question at all or the explicit question does not involve the ineffability configuration. In either case, we can identify an implicit question that is good given the context and that the example answers well.²⁸

The situation is clearly different in the *crepes* example (1), where *only* appears explicitly in the preceding question. As we have seen, the implicit question addressed by (1) cannot be identical to the background question, since reconstructing that question would require an impossible covariation between focus and question alternatives. Hence, if the implicit question reconstructed from focus must match the explicit background question — as dictated by Q-A-CONGRUENCE — ineffability is predicted. Admittedly, an answer may also target an implicit question that differs in a restricted way from the original explicit one, and we return to this possibility in Subsection 4.3. Before doing so, however, we consider some immediate predictions of the present analysis.

One prediction is that ineffability could be reproduced with (8) and (35)-(36) if we force these examples to address an explicit question parallel to (1). This prediction seems to be borne out:

- (39) a. What do rice growers do *only* to RICE?
b. #They *only* EAT rice.
- (40) a. What do you do *only* to CREDIT CARDS?
b. #We *only* ACCEPT credit cards.

(39) can no longer mean that rice growers eat rice and no other food. Similarly, association with *credit cards* is no longer possible in (40). An apparent paradox, then: association with a direct object across an accented verb is generally possible; however, once the background question establishes that the direct object should be the operator’s associate, such association becomes impossible in the answer. This puzzle is resolved by our question-based approach: the presence of *only* in the question entails that the main focus cannot lie within *only*’s scope in the answer; that would require an impossible covariation of the operator’s domain across question alternatives, just as in our ineffability example (1).

Conversely, our derivation of ineffability was based on the observation that it is the particular question in (1) that cannot be answered by either (1-a) or (1-b). It is therefore predicted that if the question in (1) were not fully determined by the context, accent placement could, in principle, be more flexible. Specifically, if *only* in (1-a) could be understood as falling under a broad focus (and therefore absent from the addressed implicit question), that same answer should become more acceptable. We

b. He *only* saw MARC at the conference.

Since *only* does not appear in the question, unless it is treated as focused within the answer, it is unclear how the question could denote a subset of the answer’s focus alternative set. By contrast, if focusing *only* makes the corresponding sentence without *only* available as an alternative, question-answer congruence can be established.

²⁸An alternative would be to assume that focusing *only* generates alternatives in which *only* is not entirely absent, but its domain is substituted with other domains. In principle, this could obviate ineffability by allowing the otherwise impossible domain variation across question alternatives. However, if this were possible, it should also apply to the *crepes* example in (1), which would undermine our explanation of ineffability. We must therefore assume that focusing *only* does not merely focus its domain.

believe that this prediction, too, is borne out in the following example:

- (41) a. In Paris?? Why are you keeping a fancy pastry chef like that in such an expensive city?
 b. Because [John only₁ eats CREPES]_F [in Paris]_{SOF₁}.

We have not provided an explanation for how the particular stress pattern in (8) and (35)-(36) comes about, and we will have to leave this question open.²⁹ What is important to us here is that the question being addressed plays a crucial role in the felicity of the relevant examples. Accounts of ineffability that attribute the problem to the structure (or the prosody) of the answer in isolation do not predict that modifying the question should make a difference. The present account does.

This line of reasoning may also extend to examples like (42), which are judged felicitous by at least some speakers:

- (42) a. John only₁ eats baguettes in PARIS_{F₁}.
 b. He also₂ only₁ eats CREPES_{F₂} [in Paris]_{SOF₁}.

Büring [2015] and Schwarzschild [2020] would both predict this example to be at least as bad as the original (1). In (42-b), *also* associates with *crepes* while *only* associates with *Paris* — a configuration which involves crossing dependencies and therefore violates the structural constraint in (3) above. The violation is even more transparent here than in the original (1), since in (42-b) the two operators whose dependencies cross are both overt.³⁰

²⁹A similar pattern is found even when *only* is omitted from (8):

- (i) a. I only have a CREDIT card.
 b. That's quite alright, we ACCEPT credit cards.

Such cases are often analyzed in terms of *local anaphoric deaccenting*, with the reduced prominence of the direct object attributed to its contextually givenness. The broad focus analysis hinted at above aligns with previous accounts of such anaphoric deaccenting [Wagner, 2012, Büring, 2019]. An alternative is to assume that anaphoric deaccenting shifts prosodic prominence from the actual (narrow) focus to an adjacent phrase. On this view, the verbal accents in (8) and (35)-(36) actually reflect F-marking on the direct objects; thus, *only* associates with the main focus, just as in ordinary, non-SOF cases. Because pitch accent has been shifted from that focus, these examples appear to instantiate the ineffability configuration. On closer inspection, however, this configuration is only apparent. A similar direction is pursued by Schwarzschild [2020], who distinguishes N-marking (newness, or non-givenness) from F-marking (focus). Schwarzschild argues that in (36), *offered* is not focused but merely N-marked, and this N-marking is responsible for the observed pitch accent:

- (ii) [They]_N [only]_N [OFFERED]_N [radiology]_F.

The broad-focus analysis has the advantage of maintaining that accent is always assigned to (some part of) the focus, without invoking a prominence-shifting mechanism. This analysis also offers a more natural account of why there should be contrast between *grow* and *eat* in (35), or between *take* and *offer* in (36): on this view, these elements form part of the focus. A remaining difficulty is that the purported focus — everything except the direct object — does not form a constituent at LF, and the solutions proposed by Wagner [2012] and Büring [2019] for cases like (i) do not straightforwardly extend to the versions involving *only* discussed here.

³⁰(42) is entirely parallel to an example discussed by Rooth [1996] following Krifka [1991]:

- (i) a. We only introduced Marilyn to JOHN Kennedy.
 b. We also only introduced SUE to John Kennedy.

Although the empirical status of such examples is not entirely clear, we tend to agree with Rooth [1996] that (i) is acceptable (for some experimental results concerning similar examples, see Beck and Vasishth [2009]).

We do not yet offer a developed account of examples like (42). However, we note that just as in (8) and (35)-(36), the question addressed by (42-b) is not necessarily fixed by context. Perhaps it is similarly possible to interpret this example as addressing an implicit question in which *only* does not appear (e.g. *Where does John eat crepes?*). In that case, ineffability should not arise. Moreover, examples similar to (42) have been reported to be degraded for some speakers and in certain contexts. This variability may be amenable to explanation under a pragmatic, context-dependent approach like the present one, but it is hard to see how a formal approach such as Beck [2006] or Büring [2015] could derive it.

4.3 Pragmatic bridging obviates ineffability

In the previous section, we assumed that when *only* appears in the background question, it cannot be omitted from the implicit question reconstructed from focus. However, shifting from an explicit question to a different, implicit one is generally possible. Consider the following question-answer pair:

- (43) a. Where are the keys?
 b. I left them on the shelf.

Evidently, there is no way of assigning focus to (43-b) that would make the denotation of (43-a) a subset of the answer’s focus alternative set. Rather, (43-b) appears to directly address a different, implicit question — namely, *Where did you leave the keys?* — and to answer (43-a) only indirectly. This new question can be inferred from the answer and is presumably accommodated by the discourse participants [Roberts, 1996, Beaver and Clark, 2008].³¹

This raises the question of why shifting the question cannot similarly rescue the *crepes* example in (1). If, as we have argued, *only* can normally be interpreted as part of the focus, and if focusing *only* can shift the original question to an *only*-less implicit one, why not do so to avoid ineffability? We argue that while this strategy can indeed obviate ineffability in some cases, the specific properties of the *crepes* example render it ineffective here.

Although question shifts are generally possible, they are not unconstrained. In particular, shifting the question does not make the original question disappear, and one cannot simply act as though it had never been asked (except in cases where the original question is insulting, rhetorical, or clearly not the intended question, which is not the case in (1)). Both the new, implicit question and its answer must relate to the original question; specifically, the result of shifting the question, potentially aided by general reasoning, must provide a good partial answer to the original question. Such a shift can be viewed as part of a *strategy* for addressing the original question (see Roberts [1996], Büring [2003]). We do not aim to provide a comprehensive theory of question shifting; however, some shifts are clearly more sensible — and therefore easier to accommodate — than others.³²

³¹An alternative analysis of such examples would substantially weaken the notion of relevance and abandon the formal Q-A-CONGRUENCE requirement entirely. For arguments against such an approach and in favor of maintaining a strong RELEVANCE condition, supplemented by the possibility of implicit question accommodation, see Benbaji-Elhadad and Doron [2025 manuscript, under review].

³²The formal definition of a strategy in Roberts [1996] is too restrictive, as a shift may be licensed even if not all answers to the new question constitute partial answers to the original one, provided that the new question moves one closer to answering the original question by yielding an indirect answer to it (see Groenendijk and Stokhof [1984] for discussion of indirect and partial answers).

Returning to (1), we can consider various ways the question might be shifted in this example. In fact, all the implicit questions derived in Section 2.2, based on various focus placements and association patterns, are potential candidates for accommodation. Most of these questions, however, were quite bad in general and thus difficult to accommodate. Other options produced questions that are generally good but do not form a reasonable strategy with the original question. For instance, placing the main focus on *Paris* — *John only eats crepes in PARIS* — shifts the question from *What does John only eat in Paris?* to *Where does John eat crepes?*. Answering this question contributes to resolving the original one only if the answer is *Paris*. Any other answer would be entirely irrelevant to the original question, making this question an unlikely candidate for accommodation.

In (44), focusing *only* alongside crepes in the answer entails a shift from Q_1 to Q_2 :

- (44) Q_1 : What does John only eat in Paris?
 Q_2 : What does John eat in Paris?
A: John [only]_F eats [CREPES]_F in Paris.

This appears to be a promising strategy. In particular, answers to Q_2 also provide indirect answers to Q_1 , since knowing what John eats in Paris narrows down the set of foods he might eat only in Paris. Q_2 may therefore count as a relatively good question in the given context. Moreover, the response A answers Q_2 well, provided that *only* in A associates with *crepes*.³³ (If *only* associated with *Paris* instead, the resulting answer would not be relevant to Q_2 : while the claim that John eats crepes in Paris is relevant, the claim that he does not eat crepes anywhere else is not.) We therefore predict that if the basic meaning obtained by associating *only* with *crepes* in A could plausibly serve as an indirect answer to the original Q_1 , ineffability should not arise.

Of course, the question arises as to why the *crepes* example cannot be rescued using this strategy. Before addressing that, let us first examine cases that we believe are indeed rescued in this way. The following examples are reported as felicitous by Beaver and Velleman [2011]:

- (45) a. What do you only eat out at a restaurant and what do you only eat at home?
b. I pretty much only eat non-EUROPEAN food out (because I don't have most of the ingredients or tools to make them in my pantry).
- (46) a. What movies have you only seen on video?
b. I've only seen JAWS on video (I've never seen it in the theatre).

Unlike the cases discussed in the previous subsection, here the background question is fixed by the context, making these examples even more closely parallel to the ineffability example in (1). Based on such examples, Beaver and Velleman join Rooth [2010] in concluding that ineffability cases are uninformative, and leave them unexplained. Given that ineffability has already been shown to follow directly from standard assumptions about the semantic content of (1) and the role of questions and answers in discourse, the remaining challenge is to explain why (45) and (46) are not equally problematic.

We argue that the felicitous readings of these examples do not, in fact, instantiate the ineffability

³³Although the proposition expressed by the answer (that John eats crepes and nothing else in Paris) is not itself a member of the denotation of Q_2 , which is $\{\text{John eats } x \text{ in Paris} \mid x \in \{\text{crepes, bananas, ...}\}\}$, it nonetheless provides a good answer to Q_2 , because it identifies a cell in the partition induced by the question (see footnote 10 and Fox [2019] for further discussion).

configuration. Instead, they involve a shift to the corresponding question without *only*, with *only* in the answer associating with the main focus. The result is then pragmatically related back to the original question. In other words, the acceptability of these examples can be attributed to the strategy we have just outlined. We find some support for this proposal in the following exchange:

- (47) a. What movies have you only seen on video?
 b. #I've only seen JAWS on video. I also saw No Country for Old Men on video, but this one I've also seen in the theatre.

We believe that the prominent reading of (47) is contradictory: the second sentence in the answer seems to conflict with the exclusive inference of the first. We take this as evidence that the associate of *only* in (46) is in fact *Jaws*, in which case ineffability is not predicted to arise: once *only*'s associate is also the main focus, the operator's domain is no longer required to vary across question alternatives.³⁴

Why, then, can the original *crepes* example not be rescued in the same way? It is possible that, for some speakers, it can. However, there are several differences between example (1) and the cases discussed by Beaver and Velleman, which may account for their differing status.

The crucial difference between (45)-(46) and (1) is that the (b)-sentences in (45) and (46) provide more plausible answers to the accommodated *only*-less questions than the answer in (1). In (45), it is not particularly odd for the speaker to say that they eat non-European food and nothing else in restaurants; and although this does not settle the question of whether they eat such food elsewhere (the sentence answers *What do you eat out?* instead of *What do you only eat out?*), general reasoning can bridge the gap between the answer and the original question: if non-European food is the only kind of food the speaker eats out, then it must also be the only kind of food they eat only out.

Similarly, in (46) it is possible that the speaker has seen *Jaws* and no other movie on video. Taking this as an answer again shifts the question from *What movies have you only seen on video?* to *What movies have you seen on video?*, but general reasoning can still relate the response back to the original question. By contrast, associating *only* with *crepes* in (1) yields the outlandish inference that on John's visits to Paris he survives on crepes and nothing else. This is presumably where we reject this parse: even if a pragmatic bridge to the original question can be established, it would not render the answer sensible. In other words, while the derivations considered in Section 2.2 violate Q-A-FELICITY by requiring the accommodation of a bad question, accommodating an *only*-less question in (1) violates Q-A-FELICITY because it produces a bad answer.³⁵

Beyond general world knowledge, two factors seem to contribute to the relative felicity of (45) and (46). First, in both cases pragmatic bridging to the original background question appears to be facilitated by the continuation provided in parentheses, which more or less explicitly spells out the inferences that *only* should have produced if it could associate with the operator's associate in the explicit question. Take (45) for example: since the speaker does not have the relevant ingredients at home, presumably they cannot make non-European food at home, so they eat it only at restaurants. Without this additional material, we believe that the relevant examples are indeed degraded. Conversely, adding such clues to the original *crepes* example may improve its acceptability, which may

³⁴It is possible that some speakers employ a repair strategy to yield another, non-contradictory reading, perhaps by resorting to covert movement (see also footnote 18). We predict that speakers who judge (47) as felicitous should be able to apply the same repair strategy more generally, including in the *crepes* example (1).

³⁵It has been noted that (1) may initially be perceived as felicitous, and become odd only upon reflection. Presumably, this is because the response is initially interpreted as an indirect answer to the background question.

explain why Rooth [2010] reports that the example improves a little when extra material is added or when *crepes* receives a rising intonation, signaling a partial answer.

Second, in (46) the *wh*-phrase is restricted specifically to movies (*what movies* rather than simply *what*). This yields a more reasonable basic interpretation, since the speaker may still have seen shows, documentaries, and so on on video, even if *Jaws* is the only movie they have seen that way. We believe a similar effect can be reproduced with the *crepes* example, at least to some extent:

- (48) a. What dessert does John only eat in PARIS?
 b. ?John only eats CREPES in Paris.

With the contextual restriction to desserts, the exhaustive inference — that John eats crepes and no other dessert in Paris — is considerably more plausible than the original inference that crepes are the only kind of food he eats in Paris. Schwarzschild [2020] likewise observes that if the question is understood as being restricted to exotic foods, understanding *only* as associating with *crepes* becomes less problematic. The example can perhaps be further improved if followed by an explanation, similar to Beaver and Velleman’s examples:

- (49) a. What dessert does John only eat in PARIS?
 b. ?John only eats CREPES in Paris (he thinks that only the French know how to make them right).

Acceptability judgments for these examples are admittedly subtle. Although it makes them challenging to work with, this is in fact predicted by the present account: the relevant computations are nontrivial, and relating the answer to the question presumably relies on slow, general-purpose reasoning. The main takeaway is that any purely formal treatment of the relevant examples stands no chance of deriving these contrasts; only an account with a pragmatic component, such as the present one, can make such fine-grained distinctions among examples that appear formally identical.

5 Conclusion

Earlier work within the tradition that views focus as essentially pragmatic [Vallduví, 1990, Dryer, 1994, Roberts, 1996] has largely concentrated on simple cases, involving either a single free focus or a single associated one. Building on Katzir [2024], the pragmatic approach comes down to two simple conditions: one linking focus to questions, and another requiring an assertion to constitute a good answer to a good question. Association with focus is indirectly restricted by (and possibly derived from) these principles, since different choices for an operator’s domain correspond to different implicit questions and result in different answers to them.

In this paper, we have extended this perspective to more complex constructions involving multiple alternative sources — two foci, or a focus and a *wh*-element. A closer examination of the alternative computations involved in such constructions allowed us to account for the otherwise surprising phenomena of focus ineffability and intervention; we have traced the problem to an operator’s domain having to covary across an alternative set.

This analysis allows us to dispense with the numerous stipulations previously proposed to categorically rule out crossing dependencies between focus operators. Unlike Beck [2006], we do not require that focus alternatives be computed compositionally and share their interpretive mechanism with question alternatives, that focus evaluation be unselective and obligatorily reset alternatives, or that *wh*-elements lack an ordinary semantic value and therefore depend on a *Q*-operator for their inter-

pretation. Nor do we need to posit otherwise unmotivated structural constraints of the kind proposed by Beck [1996] for wh-intervention or by Schwarzschild [2020] for ineffability. We may likewise forgo Büring [2015]’s complex system of prosodic domains in favor of a simpler prosodic component. As we have seen, these formal stipulations are not only unnecessary, but also empirically inadequate.

Instead, the approach pursued here relies solely on widely accepted assumptions — namely, that quantificational domains are contextual variables; that focus marking gives rise to a set of minimally different alternatives; and that any accommodated question should be reasonable given its context. This question-based approach is not only more parsimonious, but also captures contrasts overlooked by purely formal accounts like those of Beck, Büring and Schwarzschild.

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A Appendix: selected derivations

This appendix provides detailed derivations for several key examples discussed in the paper. In cases where multiple interpretations were considered in the main text, we select specific parses here and illustrate how they can be derived compositionally. For clarity, we borrow the notation from Rooth [1985, 1992] to mark focus alternative sets (although we do not commit ourselves to the compositional propagation of focus alternatives, which is central to Rooth’s proposal): we mark an expression’s ordinary interpretation with $\llbracket \cdot \rrbracket_o$, and its corresponding focus alternative set with $\llbracket \cdot \rrbracket_f$.

A.1 Focus ineffability

(1-a) #John only eats CREPES in Paris.

- $\llbracket \text{John eats crepes in Paris} \rrbracket_o = \lambda w. \text{John eats crepes in Paris in } w$

We assume a simplified semantics for *only*, according to which it negates all non-weaker alternatives to its prejacent. *only*’s quantificational domain is a contextual variable marked below with C:

- $\llbracket \text{only}_C \rrbracket_o = \lambda p \lambda w : p(w). \forall q \in C [p \not\Rightarrow q \implies \neg q(w)]$

Assuming that *only* is taken to associate with *Paris*, we set:

- $C = \{\lambda w. \text{John eats crepes in } y \text{ in } w \mid y \text{ is a place}\}$

This yields the following interpretation:

- $\llbracket \text{John only}_C \text{ eats crepes in Paris} \rrbracket_o =$
 $\lambda w : \text{John eats crepes in Paris in } w. \forall y \neq \text{Paris} [\neg(\text{John eats crepes in } y \text{ in } w)]$

The focus alternative set of the sentence, understood as the implicit question addressed by it, is obtained by replacing the focused element *crepes* by (relevant) alternative individuals:

- $\llbracket \text{John only}_C \text{ eats crepes in Paris} \rrbracket_f =$
 $\{\lambda w : \text{John eats } x \text{ in Paris in } w. \forall y \neq \text{Paris} [\neg(\text{John eats crepes in } y \text{ in } w)] \mid x \text{ is a food}\}$
 $\approx \text{‘What does John eat in Paris without eating crepes anywhere (else)?’}$

The presence of *crepes* within each focus alternative is due to the fact that the operator’s domain must be shared across alternatives (it is not focused).

A.2 Overt movement obviates ineffability

(23) CREPES₁ 1 John only eats t_1 in Paris.

This time, the operator’s domain contains a bound variable:

- $C = \{\lambda w. \text{John eats } t_1 \text{ in } y \text{ in } w \mid y \text{ is a place}\}$
- $\llbracket \text{John only}_C \text{ eats } t_1 \text{ in Paris} \rrbracket_o =$
 $\lambda w : \text{John eats } t_1 \text{ in Paris in } w. \forall y \neq \text{Paris} [\neg(\text{John eats } t_1 \text{ in } y \text{ in } w)]$

Applying Predicate Abstraction [Heim and Kratzer, 1998], we get:

- $\llbracket 1 \text{ John only}_C \text{ eats } t_1 \text{ in Paris} \rrbracket_o =$
 $\lambda x \lambda w : \text{John eats } x \text{ in Paris in } w. \forall y \neq \text{Paris} [\neg(\text{John eats } x \text{ in } y \text{ in } w)]$

crepes combines with this expression to return the following proposition:

- $\llbracket \text{crepes}_1 \text{ 1 John only}_C \text{ eats } t_1 \text{ in Paris} \rrbracket_o =$
 $\lambda w : \text{John eats crepes in Paris in } w. \forall y \neq \text{Paris} [\neg(\text{John eats crepes in } y \text{ in } w)]$

Focus alternatives are generated by replacing the constituent corresponding to *crepes* in the LF by its alternatives; the domain is thus similarly bound in each focus alternative. We get the following set of focus alternatives:

- $\llbracket \text{crepes}_1 \text{ 1 John only}_C \text{ eats } t_1 \text{ in Paris} \rrbracket_f =$
 $\lambda w : \text{John eats } x \text{ in Paris in } w. \forall y \neq \text{Paris} [\neg(\text{John eats } x \text{ in } y \text{ in } w)]$
 $\approx \text{'What does John eat in Paris and nowhere else?'}$

A.3 F-WH association

(32) ‘Libai only attended WHAT ACTIVITY?’

- $\llbracket \text{Libai attended what activity} \rrbracket_o =$
 $\{\lambda w. \text{Libai attended } x \text{ in } w \mid x \text{ is an activity}\}$

In this example, the operator associates with the wh-element, that is, its domain is formed by the wh-alternatives:

- $C = \{\lambda w. \text{Libai attended } x \text{ in } w \mid x \text{ is an activity}\}$
- $\llbracket \text{Libai only}_C \text{ attended what activity} \rrbracket_o =$
 $\{\lambda w. \text{Libai attended } x \text{ in } w. \forall x' \neq x [\neg(\text{Libai attended } x' \text{ in } w)] \mid x, x' \text{ are activities}\}$
 $\approx \text{'What activity did Libai attend without attending any other activity?'}$

A.4 Multiple associations

The following example is discussed by Krifka [1991]:

(50) (At yesterday’s party, people stayed with their first choice of drink. Bill only drank WINE, Sue only drank BEER, and...) John even only drank WATER.

In (50), both *even* and *only* associate with the focus on *water*: the sentence conveys that John drank water and nothing else (focus association with *only*), while also presupposing that it was more likely he would have chosen something other than water as his sole drink (the contribution of *even*). As Erlewine [2025] notes, this pattern poses an empirical challenge for the proposal in Beck [2006]: if the lower operator *only* resets focus alternatives introduced below it, it should make them inaccessible to the higher *even*.

In this example, *even* is arguably part of the focus — it does not contribute to the implicit question addressed by the sentence (see also Section 4.2). This makes things easier for us, so let us consider instead a slightly different version of (50), one in which both operators can plausibly be taken to participate in the implicit question:

(51) (At yesterday’s party, people stayed with their first choice of drink. Bill only drank WINE, Sue only drank BEER, and John even only drank something disgusting, but you’ll never guess

what. Let me try...)
 He even only drank [SALT WATER]_F.

The implicit question reconstructed from focus is the following:

$$(52) \quad Q = \{ \text{John even}_{C_{\text{even}}} \text{ only}_{C_{\text{only}}} \text{ drank } x \mid x \in \{ \text{wine, beer, water...} \} \}$$

Assuming that both *only* and *even* associate with the focus on *salt water*, we obtain:

$$(53) \quad \begin{aligned} \text{a. } C_{\text{only}} &= \{ \text{John drank } x \mid x \in \{ \text{wine, beer, water, ...} \} \} \\ \text{b. } C_{\text{even}} &= \{ \text{John only}_{C_{\text{only}}} \text{ drank } x \mid x \in \{ \text{wine, beer, water...} \} \} \\ &= \{ \text{John drank } x \text{ and nothing else} \mid x \in \{ \text{wine, beer, water, ...} \} \} \end{aligned}$$

Taking *only*'s and *even*'s contributions into account, the resulting question is $Q = \{ \text{John drank } x \text{ and nothing else, despite being more likely to choose something other than } x \text{ as his sole drink} \mid x \in \{ \text{wine, beer, water, ...} \} \}$. This is a good question given the context. The answer further addresses it properly, and the overall interpretation is exactly as expected.